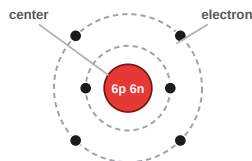


How Atoms Build Everything

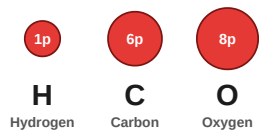
From the tiniest piece
to stuff you can touch, in 8 steps



Carbon: heavy center + light electrons

1 The atom

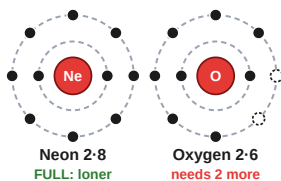
A tiny, heavy center of **protons** and **neutrons**, glued by the strong force. Light **electrons** zoom around it in layers called **shells**.



Count the protons → know the element

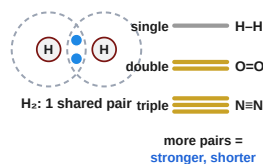
2 Protons are the ID

The number of protons decides the element. 1 = hydrogen, 6 = carbon, 8 = oxygen. It is the big number on each square of your periodic table.



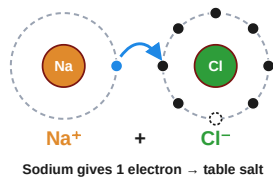
3 Atoms want a full outer shell

The outer shell wants **8** electrons, or **2** if the atom has only one shell. Full atoms (noble gases) stay alone. Everyone else is missing some.



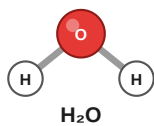
4 Sharing makes a bond

Two atoms share electrons, and **shared electrons count for both**. Each shared pair = 1 bond = 1 stick in your kit. More pairs make a stronger bond: one hand, two hands, bear hug!



5 Or: give and take

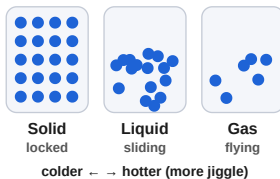
Some atoms hand an electron over instead. Sodium gives 1 to chlorine, so both are full. Now one is + and one is -, and they stick like magnets. That is table salt!



2 hydrogens + 1 oxygen, bent shape

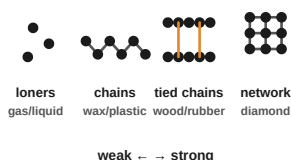
6 A molecule

A group of atoms bonded together. Water is 2 hydrogens + 1 oxygen. Its **shape** matters too: water is bent like a boomerang.



7 Molecules stick together (a little)

Separate molecules cling weakly. Heat makes them jiggle. Stickiness wins = **solid**. A tie = **liquid**. Jiggling wins = **gas**.



8 Size and connections make the stuff

Trillions of molecules make a substance (1.7 sextillion in one drop of water!). Small loners are gases and liquids. Longer chains get more solid. Tied-together chains and networks are toughest.

In one sentence: Atoms share electrons to fill their outer shells. That makes molecules, and how big and how sticky the molecules are decides whether something is a gas, a liquid or a solid.